SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence COURSE CODE: CSA1741

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks: 50**

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

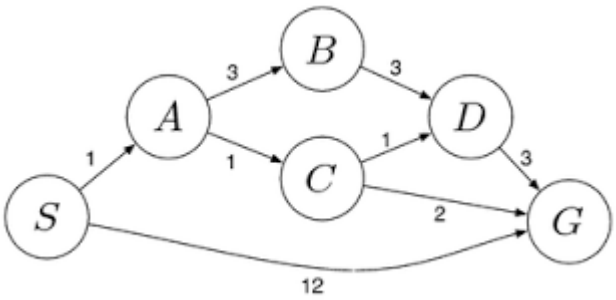
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I VISHAL.V(192419033) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: VISHAL.V

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

SOLUTIONS: -

1.Water Jug Problem

Introduction

The Water Jug Problem is a classic AI problem in which we must measure an exact amount of water using two jugs without any measuring marks. We have a 4-gallon jug and a 3-gallon jug, and we need to obtain exactly 2 gallons in the 4-gallon jug.

Steps to Solve:

- Fill the 3-gallon jug completely.
- Pour the water from the 3-gallon jug into the 4-gallon jug. The 4-gallon jug now contains 3 gallons.
- Fill the 3-gallon jug again.
- Pour water into the 4-gallon jug until it is full. Only 1 gallon can be added, leaving 2 gallons in the 3-gallon jug.
- Empty the 4-gallon jug.
- Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug.

State Table:

Step	4-gallon jug	3-gallon jug
Initial state	0	0
Fill 3-gallon jug	0	3
Pour into 4-gallon jug	3	0
Fill 3-gallon jug again	3	3
Fill 4-gallon jug	4	2
Empty 4-gallon jug	0	2
Pour remaining water	2	0

Conclusion

Using filling, pouring and emptying operations, we obtain exactly 2 gallons of water in the 4-gallon jug. This problem demonstrates how AI can solve problems by moving from one state to another.

2. Mars Rover as an Intelligent Agent

Introduction:
A Mars Rover is a robotic vehicle sent to Mars to explore the surface, collect soil and rock samples, take photographs and send information to Earth. It uses sensors and intelligent decision-making to work in an unfamiliar environment.

i. Percepts of the Agent
Percepts are the information received by the rover through its sensors.

- Images of the Martian surface.
- Distance from rocks and obstacles.
- Soil and rock composition.
- Temperature and environmental conditions.
- Battery level and location.

ii. Operating Environment
The rover operates on the surface of Mars. Its environment contains rocks, craters, dust, uneven ground and extreme temperatures. It is a partially observable, uncertain and changing environment because the rover cannot know everything about its surroundings at once.

iii. Actions of the Agent
The rover can:

- Move forward, backward and turn.
- Capture images using cameras.
- Collect soil and rock samples.
- Analyse samples using scientific instruments.
- Avoid obstacles and send data to Earth.

iv. Performance Evaluation
The rover's performance can be evaluated based on:

- Number and quality of samples collected.
- Accuracy of scientific data.
- Distance explored successfully.
- Amount of energy consumed.
- Safety of the rover and successful transmission of data.

v. Suitable Agent Architecture
A model-based, goal-based agent is suitable for a Mars Rover. The model-based agent remembers information about its surroundings, while the goal-based agent plans actions to achieve objectives such as reaching a location or collecting samples.
A utility-based component can also help select safer routes while saving energy.

Conclusion:

An intelligent Mars Rover can explore Mars safely and collect useful scientific information. Combining memory, goal-based planning and decision-making helps it work effectively in an unknown environment.

3. Eight Queens Problem

Introduction

The Eight Queens Problem is a famous AI problem in which eight queens must be placed on an 8 × 8 chessboard so that no queen attacks another queen.

Queens can attack horizontally, vertically and diagonally. Therefore, no two queens should share the same row, column or diagonal.

Problem Formulation

- Initial state: An empty chessboard.
- State: A board containing some of the eight queens.
- Actions: Place a queen in a safe position.
- Goal test: Check whether all eight queens are placed without attacking each other.
- Path cost: Each queen placement can be assigned a cost of 1.

Search Strategy

Backtracking search can be used to solve this problem.

Steps:

1. Start with the first column.
2. Place a queen in a safe row.
3. Move to the next column and place another queen.
4. If no safe position is available, remove the previous queen and try another row.
5. Repeat until all eight queens are placed safely.

Example Representation

The following is one valid solution, where each number represents the row containing the queen in that column.

Column	1	2	3	4	5	6	7	8
Queen's row	1	5	8	6	3	7	2	4

No two queens in this arrangement share a row, column or diagonal.

Conclusion

The Eight Queens Problem demonstrates how AI uses search and backtracking to find a valid solution while avoiding conflicting states.

4. OLA Cab Booking Using a Problem-Solving Agent

Introduction

An OLA cab booking application helps a customer travel from one location to another. The customer enters the pickup and destination locations and selects a cab type based on comfort, cost and preference. A goal-based problem-solving agent is suitable because it works towards the goal of reaching the destination.

Working of the Agent

1. Receive the pickup location and destination.
2. Identify the available cab types, such as Mini, Micro, Sedan and Prime.
3. Compare the estimated fare, travel time and customer preferences.
4. Select a suitable cab and route.
5. Complete the booking and help the customer reach the destination.

Pseudocode:

```
START

Input pickup_location
Input destination
Input preferred_cab_type

Find available cabs

If preferred cab is available:
    Select preferred cab
Else:
    Display available cabs

Compare fare and travel time
Confirm cab booking
Navigate to destination

STOP
```

Example

Suppose a customer wants to travel from home to college and prefers a Sedan. The agent checks whether a Sedan is available, considers the fare and confirms the booking if the customer accepts it.

Conclusion

A goal-based problem-solving agent helps customers reach their destination by selecting a suitable cab and route. Considering fare, time and comfort makes the booking process more useful.

5. Uniform Cost Search (UCS) for a Delivery Network

Introduction

Uniform Cost Search (UCS) is an uninformed search algorithm that finds the path with the lowest total cost. In a delivery network, each route has a cost based on factors such as fuel and travel time.

The objective is to find the least-cost path from the starting warehouse S to the destination warehouse G.

Given Routes and Costs:

Route	Cost
S → A	1
S → G	12
A → B	3
A → C	1
B → D	3
C → D	1
C → G	2
D → G	3

Steps Using UCS

1. Start at warehouse S with cost 0.
2. From S, explore A with cost 1 and G with cost 12.
3. Select A because it has the lowest cost. From A, explore B with total cost \(1+3=4\) and C with total cost \(1+1=2\).
4. Select C with cost 2. From C, explore D with total cost \(2+1=3\) and G with total cost \(2+2=4\).
5. Select D with cost 3. Its route to G has total cost \(3+3=6\), which is more expensive than the existing route to G with cost 4.
6. The lowest-cost goal path is therefore selected.

Cost Calculation

The best path is:

$$S \rightarrow A \rightarrow C \rightarrow G$$

Total cost:

$$1 + 1 + 2 = \boxed{4}$$

Comparison of Possible Paths

Path	Total cost
S → G	12
S → A → B → D → G	10
S → A → C → D → G	6
S → A → C → G	4

Conclusion

Using Uniform Cost Search, the delivery company can transport packages from warehouse S to warehouse G at a minimum cost of 4 units by following the path S → A → C → G. UCS is useful because it considers the total route cost rather than simply choosing the nearest next warehouse.